Fully AWS Hosted Blog webapp

1. Create a RDS Database to store the Blogs:

DB instance identifier
Type a name for your DB instance. The name must be unique across all DB instances owned by your AWS account in the current AWS Region.

blogDB

The DB instance identifier is case-insensitive, but is stored as all lowercase (as in "mydbinstance"). Constraints: 1 to 63 alphanumeric characters or hyphens. First char
a hyphen.

Master username Info
Type a login ID for the master user of your DB instance.

admin

1 to 16 alphanumeric characters. The first character must be a letter.

Credentials management
You can use AWS Secrets Manager or manage your master user credentials.

Managed in AWS Secrets Manager - *most secure*
RDS generates a password for you and manages it throughout its lifecycle using AWS Secrets Manager.

Self managed
Create your own password or have R

Auto generate password
Amazon RDS can generate a password for you, or you can specify your own password.

Master password Info

••••••••

Password strength Weak

Minimum constraints: At least 8 printable ASCII characters. Can't contain any of the following symbols: / ' " @

Confirm master password Info

blogdb123

2. Now Create a S3 Bucket for Frontend files:

General purpose buckets (3) Info

Copy ARN | Empty | Delete | Create bucket

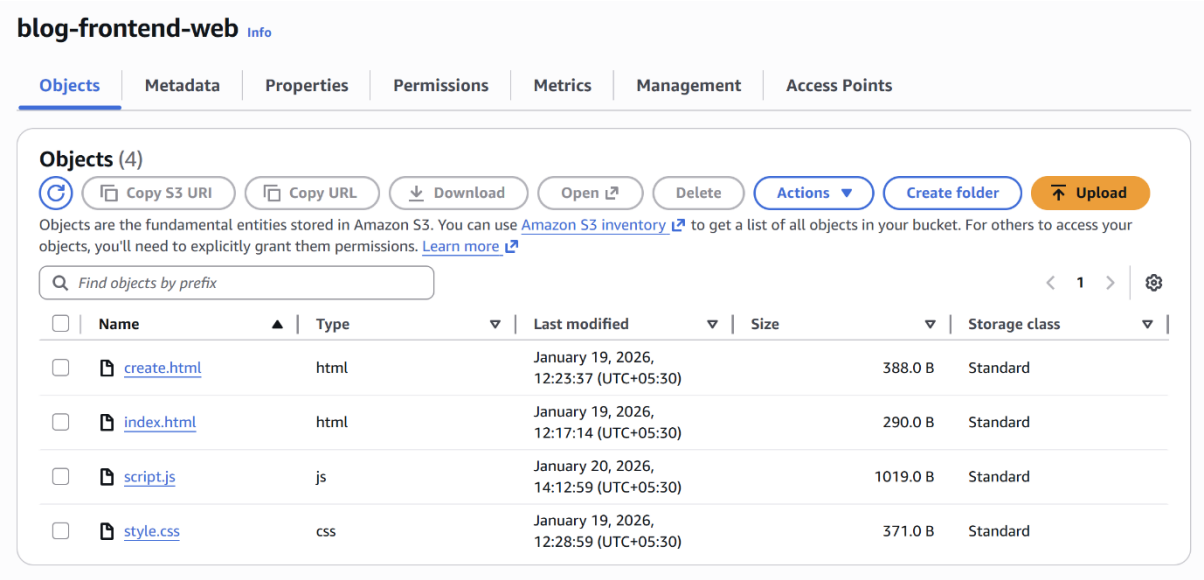
Buckets are containers for data stored in S3.

Find buckets by name

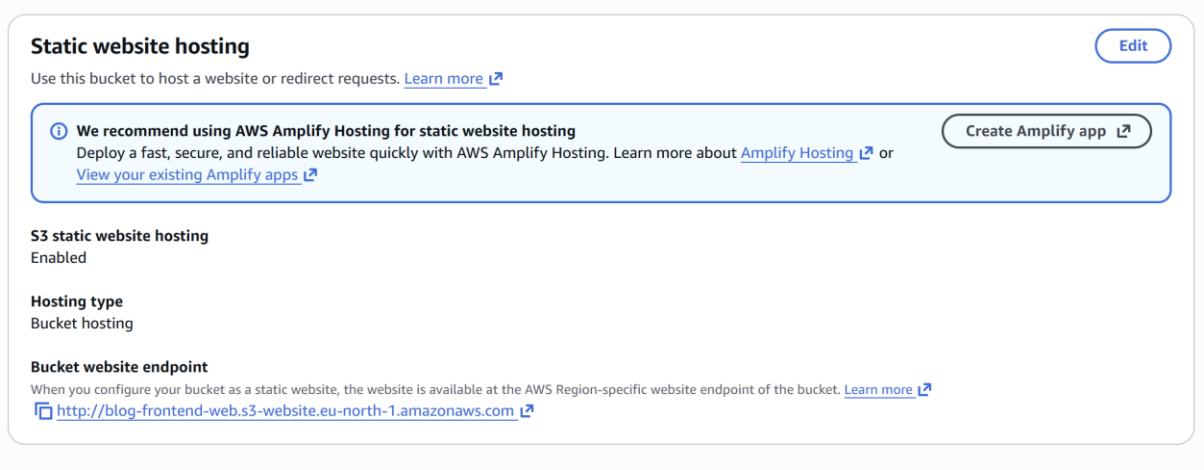
1

Name	AWS Region	Creation date
blog-frontend-web	Europe (Stockholm) eu-north-1	January 19, 2026, 12:08:18 (UTC+05:30)
event-driven-data-bucket-1	Asia Pacific (Mumbai) ap-south-1	January 5, 2026, 21:52:07 (UTC+05:30)
hasrat-resume	Asia Pacific (Mumbai) ap-south-1	December 17, 2025, 19:58:31 (UTC+05:30)

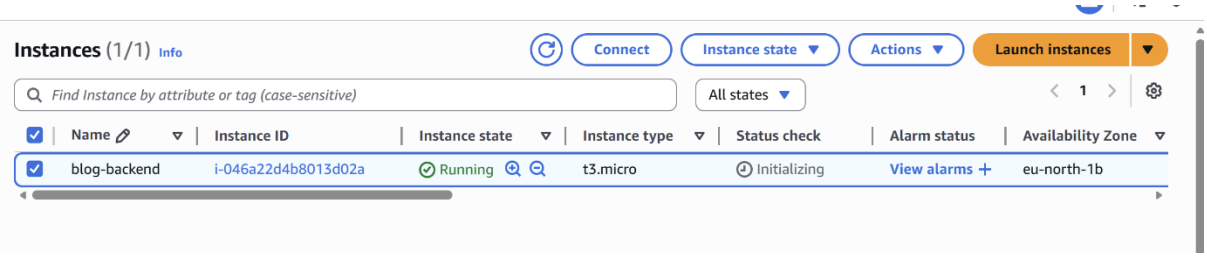
3. Upload the Frontend files inside the S3 bucket:



4. Enable static Web Hosting inside the bicket Properties Tab:



5. Create one EC2 intance for hosting backend:



6. Connect to EC2 via SSH:

```
PS C:\Windows\system32> ssh -i C:\Users\hasan\Downloads\blog-backend.pem ec2-user@ec2-13-60-29-97.eu-north-1.compute.ama
zonaws.com
The authenticity of host 'ec2-13-60-29-97.eu-north-1.compute.amazonaws.com (13.60.29.97)' can't be established.
ED25519 key fingerprint is SHA256:MkomsYoVbbjcDAqhg6vmVue0dycl/xZ01PLG73H57Gw.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ec2-13-60-29-97.eu-north-1.compute.amazonaws.com' (ED25519) to the list of known hosts.
       #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
[ec2-user@ip-172-31-41-13 ~]$
```

7. Install Node.js:

```
[ec2-user@ip-172-31-41-13 ~]$ sudo yum install nodejs
Amazon Linux 2023 Kernel Livepatch repository                                    239 kB/s |  31 kB     00:00
Dependencies resolved.
================================================================================================================
 Package                    Architecture     Version                                    Repository          Size
================================================================================================================
Installing:
 nodejs                     x86_64           1:18.20.8-1.amzn2023.0.2                   amazonlinux         13 M
Installing dependencies:
 libbrotli                  x86_64           1.0.9-4.amzn2023.0.2                       amazonlinux        315 k
 nodejs-libs                x86_64           1:18.20.8-1.amzn2023.0.2                   amazonlinux         14 M
Installing weak dependencies:
 nodejs-docs                noarch           1:18.20.8-1.amzn2023.0.2                   amazonlinux        7.8 M
 nodejs-full-i18n           x86_64           1:18.20.8-1.amzn2023.0.2                   amazonlinux        8.4 M
 nodejs-npm                 x86_64           1:10.8.2-1.18.20.8.1.amzn2023.0.2          amazonlinux        1.9 M

Transaction Summary
================================================================================================================
Install  6 Packages
```

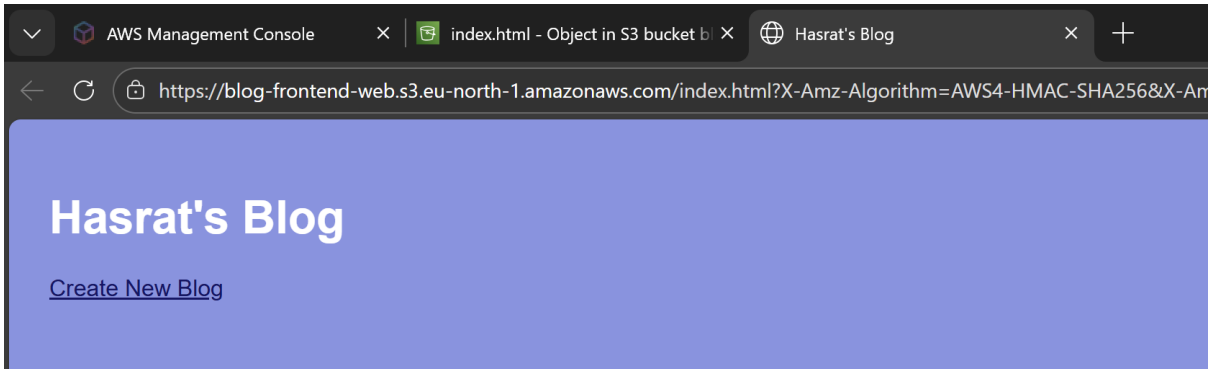
8. Make directory named Blog-Backend And put Backend Code inside the directory with NANO command:

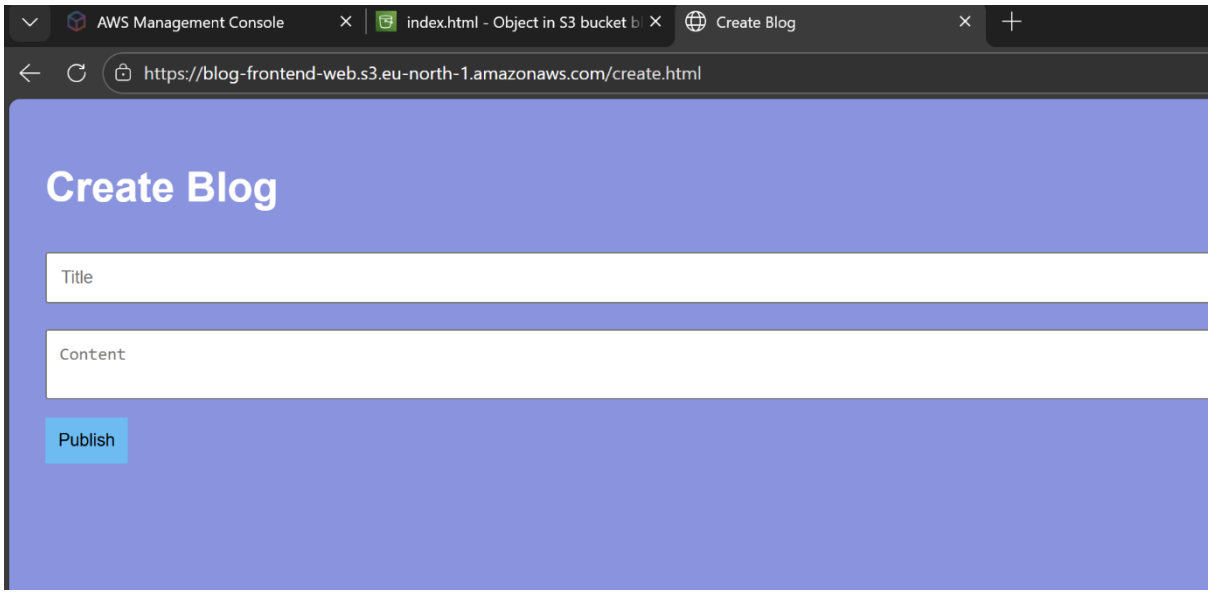
```
[ec2-user@ip-172-31-41-13 ~]$ mkdir blog-backend
[ec2-user@ip-172-31-41-13 ~]$ ls
blog-backend
[ec2-user@ip-172-31-41-13 ~]$ cd blog-backend
[ec2-user@ip-172-31-41-13 blog-backend]$
```

9. Start the Backend:

```
MySQL [blogdb]> exit
Bye
[ec2-user@ip-10-0-27-172 blog-backend]$
[ec2-user@ip-10-0-27-172 blog-backend]$ node server.js
Backend running on port 5000
Connected to RDS MySQL
```

10. Now go to frontend Blog app and Create New blog:





11. To check if the Backend is running check if the data is stored in the RDS Database :
    Go to Ec2 Instance and Install MariaDB :

```
[ec2-user@ip-172-31-41-13 blog-backend]$ sudo yum install mariadb105 -y
Last metadata expiration check: 0:16:55 ago on Wed Feb 11 05:40:53 2026.
Dependencies resolved.
================================================================================================================
 Package                          Architecture     Version                          Repository          Size
================================================================================================================
Installing:
 mariadb105                       x86_64           3:10.5.29-1.amzn2023.0.1         amazonlinux        1.5 M
Installing dependencies:
 mariadb-connector-c              x86_64           3.3.10-1.amzn2023.0.1            amazonlinux        211 k
 mariadb-connector-c-config       noarch           3.3.10-1.amzn2023.0.1            amazonlinux        9.9 k
 mariadb105-common                x86_64           3:10.5.29-1.amzn2023.0.1         amazonlinux         28 k
 perl-Sys-Hostname                x86_64           1.23-477.amzn2023.0.7            amazonlinux         16 k

Transaction Summary
================================================================================================================
Install  5 Packages
```

12. Now Connect the Ec2 with with our RDS Database to run Commands:

```
[ec2-user@ip-10-0-27-172 blog-backend]$ mysql -h blog-db.c1kg4ws6y821.eu-north-1.rds.amazonaws.com -u admin -p blogdb
Enter password:
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MySQL connection id is 4764
Server version: 8.4.6 Source distribution

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MySQL [blogdb]>
```

13. Run SQL Commands to check the Stored Blog app DATA:

```
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MySQL connection id is 4764
Server version: 8.4.6 Source distribution

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MySQL [blogdb]> SHOW DATABASES;
+--------------------+
| Database           |
+--------------------+
| blogdb             |
| information_schema |
| mysql              |
| performance_schema |
| sys                |
+--------------------+
5 rows in set (0.002 sec)

MySQL [blogdb]> USE blogdb;
Database changed
MySQL [blogdb]> SHOW TABLES;
+------------------+
| Tables_in_blogdb |
+------------------+
| blogs            |
+------------------+
1 row in set (0.002 sec)

MySQL [blogdb]> SELECT * FROM blogs;
```

```
MySQL [(none)]> USE blogdb
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
MySQL [blogdb]> SELECT * FROM blogs;
+----+-------+---------+---------------------+
| id | title | content | created_at          |
+----+-------+---------+---------------------+
|  1 | evev  | wdcedce | 2026-01-20 09:31:29 |
|  2 | evev  | wdcedce | 2026-01-20 09:31:29 |
|  3 | evev  | wdcedce | 2026-01-20 09:31:29 |
|  4 | evev  | wdcedce | 2026-01-20 09:31:29 |
|  5 | evev  | wdcedce | 2026-01-20 09:31:30 |
|  6 | evev  | wdcedce | 2026-01-20 09:31:30 |
+----+-------+---------+---------------------+
6 rows in set (0.001 sec)

MySQL [blogdb]>
```